

MVLU COLLEGE

Subject:-Cyber & Information Security

Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

Code:-

```
T098 CIS P3.py - C:/Users/mvluc/Desktop/T098/T098 CIS P3.py (3.14.6)
File Edit Format Run Options Window Help

import hmac
import hashlib

# Secret key
secret_key = b"cybersecurity"

# Original message
message = input("Enter the message: ").encode()

# Generate MAC
mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:")
print(mac)

# Verification
received_message = input("\nEnter message for verification: ").encode()
received_mac = input("Enter received MAC: ")

new_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()

if hmac.compare_digest(new_mac, received_mac):
    print("\nMAC Verification Successful.")
    print("Message is Authentic and Unchanged.")
else:
    print("\nMAC Verification Failed.")
    print("Message has been Modified or MAC is Incorrect.")
```

```
T098 CIS P3.py - IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/mvluc/Desktop/T098/T098 CIS P3.py =====
Enter the message: T098_Vikas

Generated MAC:
317958b9a92265e4981c44d2a55fd5a98a7b8e34822460f43ffc8cc0bca99bec

Enter message for verification: T098_Vikas
Enter received MAC: 317958b9a92265e4981c44d2a55fd5a98a7b8e34822460f43ffc8cc0bca99bec

MAC Verification Successful.
Message is Authentic and Unchanged.
```